SQL Fundamentals: SELECT & Filtering

01. Selecting & Sorting

1. Retrieve all columns from the employees table.

```
SELECT *
FROM employees;
```

id	first_name	last_name	department	salary	hire_date	city
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

2. Find all unique departments.

```
SELECT DISTINCT department
FROM employees;
```

department
IT
HR
Finance
Marketing

3. Retrive first and last names ordered by salary descending.

```
SELECT    first_name,
          last_name
FROM      employees
ORDER BY  salary DESC;
```

first_name	last_name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

4. Retrieve the top 3 highest-paid employees.

```
SELECT id,
        first_name,
        last_name,
        salary
FROM employees
ORDER BY salary DESC
LIMIT 3;
```

id	first_name	last_name	salary
5	Eva	Naidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

02. Filtering with WHERE

5. Find employees in the IT department.

```
SELECT id,
        first_name,
        last_name,
        department
FROM employees
WHERE department = 'IT';
```

id	first-name	last-name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

6. Find employees in Finance with salary > 60 000.

```
SELECT id,
        first_name,
        last_name,
        department,
        salary
FROM    employees
WHERE department = 'Finance' AND salary > 60 000;
```

id	first-name	last-name	department	salary
3	Cathy	Zulu	Finance	65000

7. Find employees in HR or Marketing.

```
SELECT id,
        first_name,
        last_name,
        department
FROM  employees
WHERE department = 'HR' OR department = 'Marketing';
```

id	first-name	last-name	department
2	Brian	Lee	HR
4	David	Motsena	Marketing

8. Find employees not in IT.

```
SELECT id,
        first-name,
        last-name,
        department
FROM employees
WHERE department <> 'IT';
```

id	first-name	last-name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Makoena	Marketing

9. Find employees in IT, HR, or Finance using IN.

```
SELECT id,
        first-name,
        last-name,
        department
FROM employees
WHERE department IN('IT', 'HR', 'Finance');
```

id	first-name	last-name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

10. Find employees in IT with salary > 65000 and city Johannesburg

```
SELECT id,
        first_name,
        last_name,
        department,
        salary,
        city
FROM
FROM    employees
WHERE department = 'IT' AND salary > 65000 AND city = 'Johannesburg';
```

id	first_name	last_name	department	salary	city
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Naidoo	IT	72000	Johannesburg